Lesson 5: Sufficient Statistics

BSTA 551: Statistical Inference

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Lesson 5: Sufficient Statistics

Review: Where We Left Off

Key concepts from Lessons 1-4:

- **Point estimation:** $\hat{\theta}$ estimates unknown parameter θ
- **Bias and MSE:** $\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta, \text{MSE} = \text{Var} + \text{Bias}^2$
- **MVUE:** Unbiased estimator with minimum variance
- **Fisher Information:** $I(\theta) = E[-\ell''(\theta)]$
- **Cramér-Rao Lower Bound:** $\text{Var}(T) \geq 1/I_n(\theta)$
- **MLE and MME:** Systematic methods for constructing estimators

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Today's Goals:

- Understand sufficient statistics: data reduction without information loss
- Learn the Neyman Factorization Theorem
- Connect sufficiency to finding MVUEs via Rao-Blackwell
- Define completeness and its role in optimal estimation

Motivating Question: Data Reduction

Suppose we observe $X_1, \ldots, X_n$ from a distribution with unknown θ.

Question: Can we summarize the data with a simpler statistic $T(X_1, \ldots, X_n)$ without losing information about θ?

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If a statistic captures **all** the information about θ contained in the data, we call it **sufficient** for θ.

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Why this matters:

- Simplifies analysis and computation
- Guides us to optimal estimators
- Fundamental for understanding MVUEs

The Concept of Sufficiency

Intuition: What Does "Sufficient" Mean?

Imagine you're a consulting statistician. A colleague ran an experiment and wants your help estimating θ.

Scenario 1: They give you all raw data: $x_1 = 1, x_2 = 0, x_3 = 3$

Scenario 2: They only tell you the total: $T = x_1 + x_2 + x_3 = 4$

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Question: Does Scenario 1 give you more information about θ than Scenario 2?

If the answer is **no**—if knowing T tells you everything about θ—then T is **sufficient** for θ.

Example: Poisson Defects (Devore 7.25)

Setup: The number of major defects on new vehicles follows $\mathrm{Poisson}(\mu)$. We observe $n = 3$ vehicles with defects $X_1 = 1, X_2 = 0, X_3 = 3$.

Key Question: Would knowing only $T = \sum X_i = 4$ provide the same information about μ as knowing all individual X_i's?

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Consider: Given that $T = 4$, what's the conditional probability of $(X_1, X_2, X_3) = (2, 1, 1)$?

$$P(X_1 = 2, X_2 = 1, X_3 = 1 | T = 4) = \frac{\frac{e^{-\mu}\mu^2}{2!} \cdot \frac{e^{-\mu}\mu^1}{1!} \cdot \frac{e^{-\mu}\mu^1}{1!}}{\frac{e^{-3\mu}(3\mu)^4}{4!}} = \frac{4}{27}$$

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Critical Observation

The μ terms **cancel out**! The conditional probability doesn't depend on μ.

Formal Definition of Sufficiency

Definition (Devore 7.3)

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Intuition

T is sufficient if it “captures” all the information in the data relevant to θ. It’s sufficient to know T; the raw data adds nothing more.

The Neyman Factorization Theorem

Finding Sufficient Statistics: The Factorization Theorem

Computing conditional distributions is difficult. The **Neyman Factorization Theorem** provides an elegant shortcut.

Theorem (Neyman-Fisher Factorization)

Let $f(x_1, \ldots, x_n; \theta)$ denote the joint pmf or pdf. Then $T = t(X_1, \ldots, X_n)$ is **sufficient for** θ if and only if:

$$f(x_1, \ldots, x_n; \theta) = g(t(x_1, \ldots, x_n); \theta) \cdot h(x_1, \ldots, x_n)$$

where:

- g depends on the data **only through** T and involves θ
- h depends on the data but **not on** θ

Understanding the Factorization

$$f(x_1, \ldots, x_n; \theta) = \underbrace{g(T; \theta)}_{\text{Involves } \theta \text{ and data through } T} \cdot \underbrace{h(x_1, \ldots, x_n)}_{\text{Depends on data, not } \theta}$$

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Strategy for Finding Sufficient Statistics

1. Write down the joint pmf/pdf
2. Identify the part that involves θ
3. Factor it so θ appears only with a function of the data
4. That function is your sufficient statistic!

Example 1: Poisson Distribution

Setup: $X_1, \ldots, X_n \overset{iid}{\sim} \text{Poisson}(\mu)$

Joint pmf:

$$f(x_1, \ldots, x_n; \mu) = \prod_{i=1}^{n} \frac{e^{-\mu} \mu^{x_i}}{x_i!} = \frac{e^{-n\mu} \mu^{\sum x_i}}{\prod_{i=1}^{n} x_i!}$$

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$$= \underbrace{e^{-n\mu}\mu^{t}}_{g(t;\mu)} \cdot \underbrace{\frac{1}{\prod_{i=1}^{n} x_i!}}_{h(x_1,\ldots,x_n)}$$

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Result

$T = \sum_{i=1}^{n} X_i$ is **sufficient for** μ.

Example 2: Uniform Distribution (Devore 7.27)

Setup: $X_1, \ldots, X_n \overset{iid}{\sim} \text{Uniform}[0, \theta]$

Joint pdf:

$$f(x_1, \ldots, x_n; \theta) = \frac{1}{\theta^n} \cdot I(\max(x_i) \leq \theta) \cdot I(\min(x_i) \geq 0)$$

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> **Result**
>
> $T = \max(X_1, \ldots, X_n)$ is **sufficient for** θ.
>
> Note: $\bar{X}$ is NOT sufficient here!

Why Isn't $\bar{X}$ Sufficient for Uniform?

```r
# Two samples with same mean but different maxima
data_A <- c(2, 4, 6, 8)  # mean = 5, max = 8
data_B <- c(1, 4, 5, 10) # mean = 5, max = 10

cat("Dataset A: mean =", mean(data_A), ", max =", max(data_A), "\n")
```

```
Dataset A: mean = 5 , max = 8
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```r
cat("Dataset B: mean =", mean(data_B), ", max =", max(data_B), "\n")
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Dataset B: mean = 5 , max = 10
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These have the same $\bar{X} = 5$, but:

- Dataset A tells us $\theta \geq 8$
- Dataset B tells us $\theta \geq 10$

Knowing only $\bar{X}$ loses information about θ!

Jointly Sufficient Statistics

When there are **multiple parameters**, we need **multiple sufficient statistics**.

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Example: Normal(μ, σ^2) with both unknown:

$$f = \frac{1}{\sigma^n} \exp\left(-\frac{\sum x_i^2 - 2\mu \sum x_i + n\mu^2}{2\sigma^2}\right) \cdot \left(\frac{1}{2\pi}\right)^{n/2}$$

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Result

$(\bar{X}, S^2)$ or equivalently $(\sum X_i, \sum X_i^2)$ are **jointly sufficient** for (μ, σ^2).

Summary: Sufficient Statistics for Common Distributions

Distribution	Parameter(s)	Sufficient Statistic(s)
Bernoulli(p)	p	$\sum X_i$
Poisson(μ)	μ	$\sum X_i$
Exponential(λ)	λ	$\sum X_i$
Normal(μ, known σ^2)	μ	$\bar{X}$
Normal(μ, σ^2)	(μ, σ^2)	$(\bar{X}, S^2)$
Uniform$[0, \theta]$	θ	$\max(X_i)$

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Practical Criterion (Devore 7.74)

T is minimal sufficient if the likelihood ratio $f(\mathbf{x};\theta)/f(\mathbf{y};\theta)$ doesn't depend on θ **if and only if** $t(\mathbf{x}) = t(\mathbf{y})$.

The Rao-Blackwell Theorem

Why Sufficiency Matters for Estimation

Key Question: How do we use sufficient statistics to find better estimators?

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Rao-Blackwell Theorem

If T is sufficient for θ and U is any unbiased estimator for $h(\theta)$, then:

$$U^* = E(U|T)$$

satisfies:

1. U^* is unbiased for $h(\theta)$
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In words: Conditioning on sufficient statistics **never increases variance** and often **decreases it**.

Proof of Rao-Blackwell

Step 1: U^* doesn't depend on θ (since T is sufficient)

Step 2: By Law of Total Expectation:

$$E(U^*) = E[E(U|T)] = E(U) = h(\theta)\checkmark$$

Step 3: By Law of Total Variance:

$$\text{Var}(U) = \text{Var}(E[U|T]) + E[\text{Var}(U|T)] = \text{Var}(U^*) + E[\text{Var}(U|T)]$$

Since $E[\text{Var}(U|T)] \geq 0$, we have $\text{Var}(U) \geq \text{Var}(U^*)$ ✓

Example: Poisson Zero-Defect Probability (Devore 7.30)

Problem: Estimate $e^{-\mu} = P(\text{no defects})$ for $X_i \sim \text{Poisson}(\mu)$.

Crude estimator: $U = I(X_1 = 0)$ is unbiased since $E(U) = P(X_1 = 0) = e^{-\mu}$

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Apply Rao-Blackwell: With sufficient statistic $T = \sum X_i$:

$$U^* = P(X_1 = 0|T = t) = \frac{P(X_1 = 0) \cdot P(\sum_{i=2}^{n} X_i = t)}{P(T = t)} = \left(1 - \frac{1}{n}\right)^t$$

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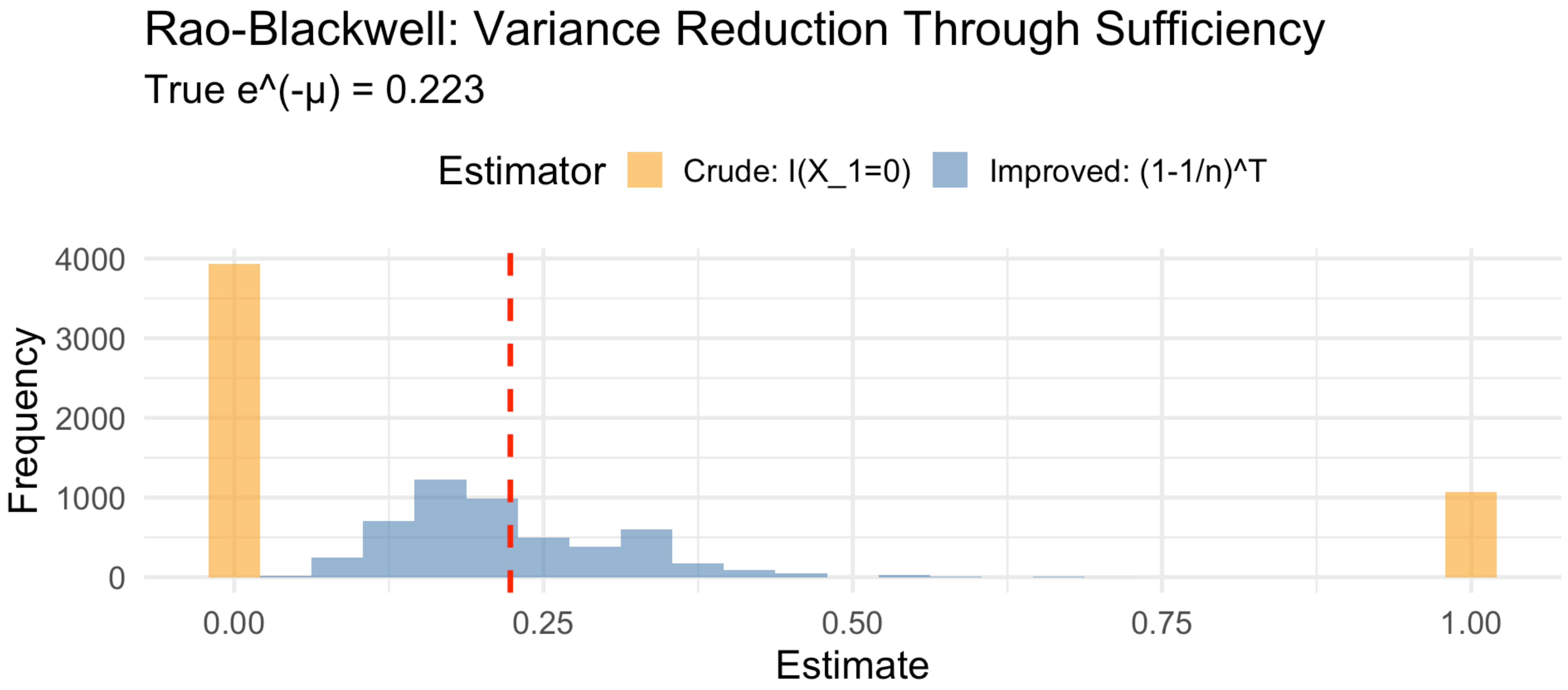
Improved Estimator

$U^* = (1 - \frac{1}{n})^T$ is unbiased with smaller variance!

Simulation: Rao-Blackwell Improvement

▶ Code



Key Implication

Corollary

Any estimator that is **not** based purely on sufficient statistics is necessarily **inferior** to some other estimator.

MLEs and Sufficient Statistics

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If T is sufficient for θ, then the MLE $\hat{\theta}$ can always be expressed as a function of T alone.

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If T is sufficient for θ, then the MLE $\hat{\theta}$ can always be expressed as a function of T alone.

Proof: By the factorization theorem:

$$L(\theta) = f(\mathbf{x}; \theta) = g(T; \theta) \cdot h(\mathbf{x})$$

Maximizing $L(\theta)$ is equivalent to maximizing $g(T; \theta)$ since $h(\mathbf{x})$ doesn't involve θ.

Therefore $\hat{\theta}$ depends on the data **only through** T.

Example: MLE Uses the Sufficient Statistic

Poisson(μ):

- Sufficient statistic: $T = \sum X_i$
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Normal(μ, σ^2)**:**

- Sufficient statistics: $(\sum X_i, \sum X_i^2)$ or equivalently $(\bar{X}, S^2)$
- MLEs: $\hat{\mu} = \bar{X}, \hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$ ✓ (functions of sufficient statistics)

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Uniform$[0, \theta]$**:**

- Sufficient statistic: $T = \max(X_i)$
- MLE: $\hat{\theta} = \max(X_i) = T$ ✓ (is the sufficient statistic itself!)

Are MLEs Themselves Sufficient?

Important distinction:

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⚠ **MLEs ≠ Sufficient Statistics**

The MLE $\hat{\theta}$ is sufficient **only if** $\hat{\theta}$ and the sufficient statistic T determine each other (one-to-one correspondence).

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Example where MLE IS sufficient:

- Uniform$[0, \theta]$: MLE = $\max(X_i)$ = sufficient statistic ✓

Example where MLE is NOT sufficient:

- Normal(μ, σ^2): MLEs are $(\hat{\mu}, \hat{\sigma}^2)$, which ARE jointly sufficient
- But if we only want $\hat{\mu} = \bar{X}$, this alone is not sufficient for both parameters

Why This Matters

Practical Implications

1. **MLEs are efficient:** They automatically use all available information about θ
2. **Finding MLEs:** Once you identify sufficient statistics, you only need to maximize $g(T; \theta)$—often simpler!
3. **Asymptotic optimality:** MLEs achieve the Cramér-Rao bound asymptotically, partly because they use sufficient statistics

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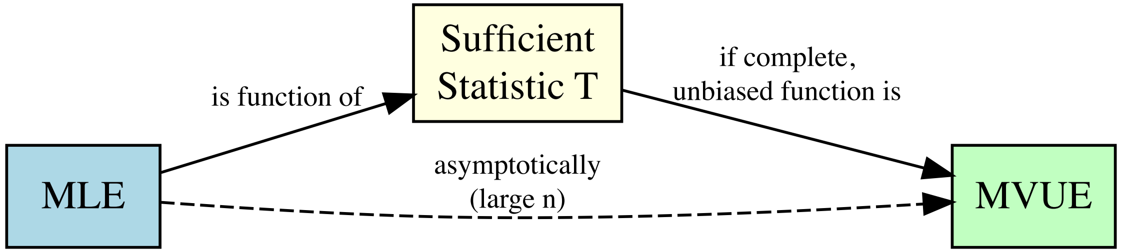
3. **Asymptotic optimality:** MLEs achieve the Cramér-Rao bound asymptotically, partly because they use sufficient statistics

Bottom line: Sufficiency explains *why* MLEs have such good properties—they never “waste” information in the data.

Connecting MLE, Sufficiency, and MVUE



- MLE is always a function of sufficient statistics
- If the sufficient statistic is complete, unbiased functions of it are MVUEs
- For large n, MLEs are approximately MVUEs (asymptotic efficiency)

Completeness

The Final Piece: Completeness

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Definition: Complete Statistic

A statistic T is **complete** if for any function g:

$$E_\theta[g(T)] = 0 \text{ for all } \theta \implies P_\theta(g(T) = 0) = 1 \text{ for all } \theta$$

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Intuition: The only unbiased estimator of zero based on T is the zero function itself.

Lehmann-Scheffé Theorem

Theorem (Lehmann-Scheffé)

If T is a **complete sufficient statistic** and $U = h(T)$ is unbiased for $\tau(\theta)$, then U is the **unique MVUE** of $\tau(\theta)$.

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Key fact: For exponential family distributions (Normal, Poisson, Exponential, Binomial, Gamma), the natural sufficient statistics are typically **complete**.

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Key fact: For exponential family distributions (Normal, Poisson, Exponential, Binomial, Gamma), the natural sufficient statistics are typically **complete**.

The Power of Completeness + Sufficiency

Once we find a complete sufficient statistic, any unbiased function of it is automatically the MVUE!

The Road to MVUE: Summary

Method 1: Via Sufficiency

1. Find sufficient statistic T (Factorization Theorem)
2. Check completeness (exponential family → typically complete)
3. Find unbiased function of T → **MVUE** (Lehmann-Scheffé)

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2. Check completeness (exponential family → typically complete)
3. Find unbiased function of T → **MVUE** (Lehmann-Scheffé)

Method 2: Via Cramér-Rao

1. Compute Fisher Information $I(\theta)$
2. Find CRLB = $1/[nI(\theta)]$
3. Find estimator achieving CRLB → **MVUE**

Summary

Lesson 5: Key Takeaways

1. **Sufficiency:** T captures all information about θ in the data
2. **Neyman Factorization:** $f(\mathbf{x};\theta) = g(T;\theta)\cdot h(\mathbf{x})$ identifies sufficient statistics
3. **Rao-Blackwell:** Conditioning on sufficient statistics improves estimators
4. **Completeness + Sufficiency → Unique MVUE** (Lehmann-Scheffé)

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2. **Neyman Factorization:** $f(\mathbf{x};\theta) = g(T;\theta)\cdot h(\mathbf{x})$ identifies sufficient statistics
3. **Rao-Blackwell:** Conditioning on sufficient statistics improves estimators
4. **Completeness + Sufficiency → Unique MVUE** (Lehmann-Scheffé)

Distribution	Sufficient Statistic	Complete?
Poisson(μ)	$\sum X_i$	Yes
Exponential(λ)	$\sum X_i$	Yes
Normal(μ, σ^2)	$(\bar{X}, S^2)$	Yes
Uniform$[0, \theta]$	$\max(X_i)$	Yes

References

- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Chapter 7.3
- Chihara and Hesterberg. *Mathematical Statistics with Resampling and R* (Wiley). Chapter 6.

Questions?

Thank you!